

HW8: Individual Effort Report

Server Design

I. Overview of Major Components and Packages

1. Server.java (Servlet)

- Handles HTTP requests (GET/POST).
- Interacts with the MySQL database.
- Publishes review events to RabbitMQ for asynchronous processing.

2. AlbumInfo.java

- Model class representing album metadata.
- Used as a data transfer object (DTO) for serializing/deserializing album data (via Gson).

3. ReviewConsumer.java

- Consumes messages from RabbitMQ (likeQueue).
- Parses and stores user reviews into the MySQL album_reviews table.

II. Data Flow and Relationships

1. Album Upload

- Stores the album into the albums table in MySQL.
- Returns AlbumInfo as a JSON response.

2. Review Upload

- Publishes it to RabbitMQ queue likeQueue.

3. Review Consumption (RabbitMQ Consumer)

- On message receive, inserts or updates the review in the MySQL album_reviews table.

Results

The throughput has been steadily kept at around 2000 in each experiment, except it jumped to around 2500 in 30 30 2, meaning that the system may have some room to grow, it could be due to queued tasks can be fed more consistently or more parallelism fills in idle CPU/network gaps,

or simple the network behaves better at that time. The sure thing right now is that the system may still have room for handling more threads.

```
server [~/Desktop/6650/CS6650/8/server] - Server.java
client - Main.java

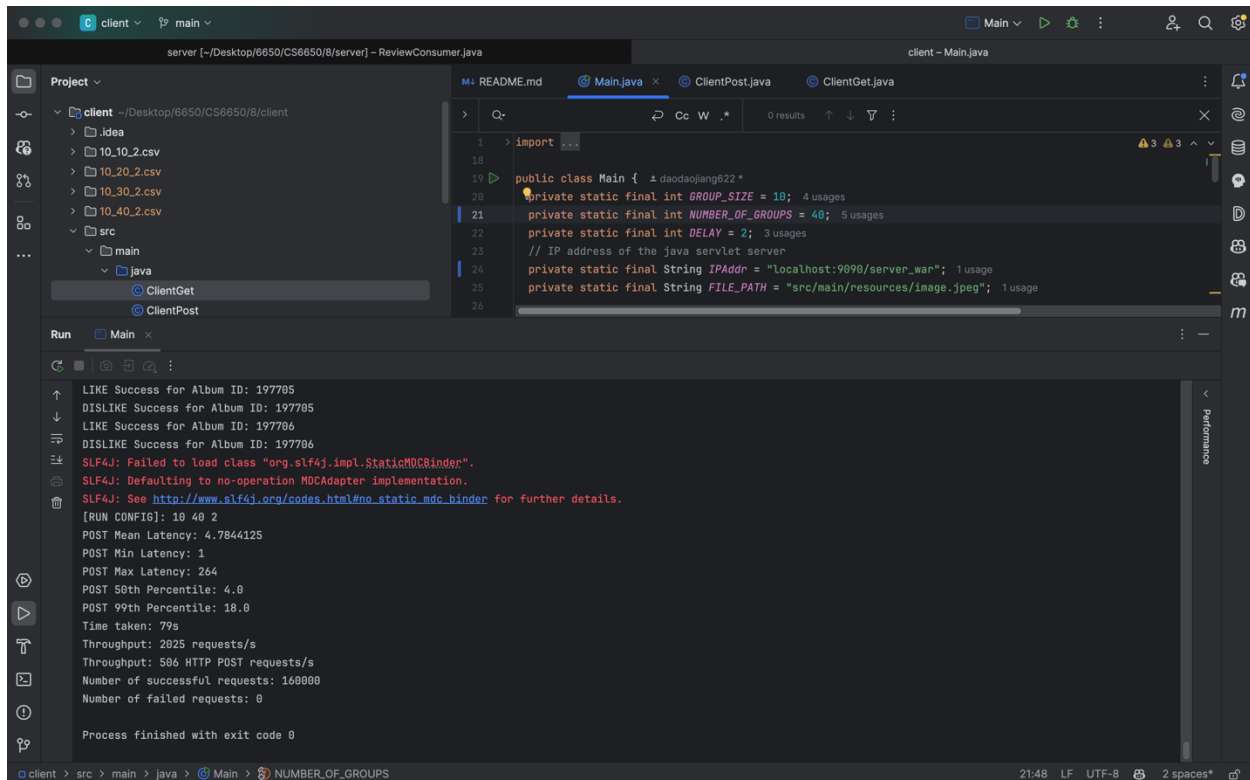
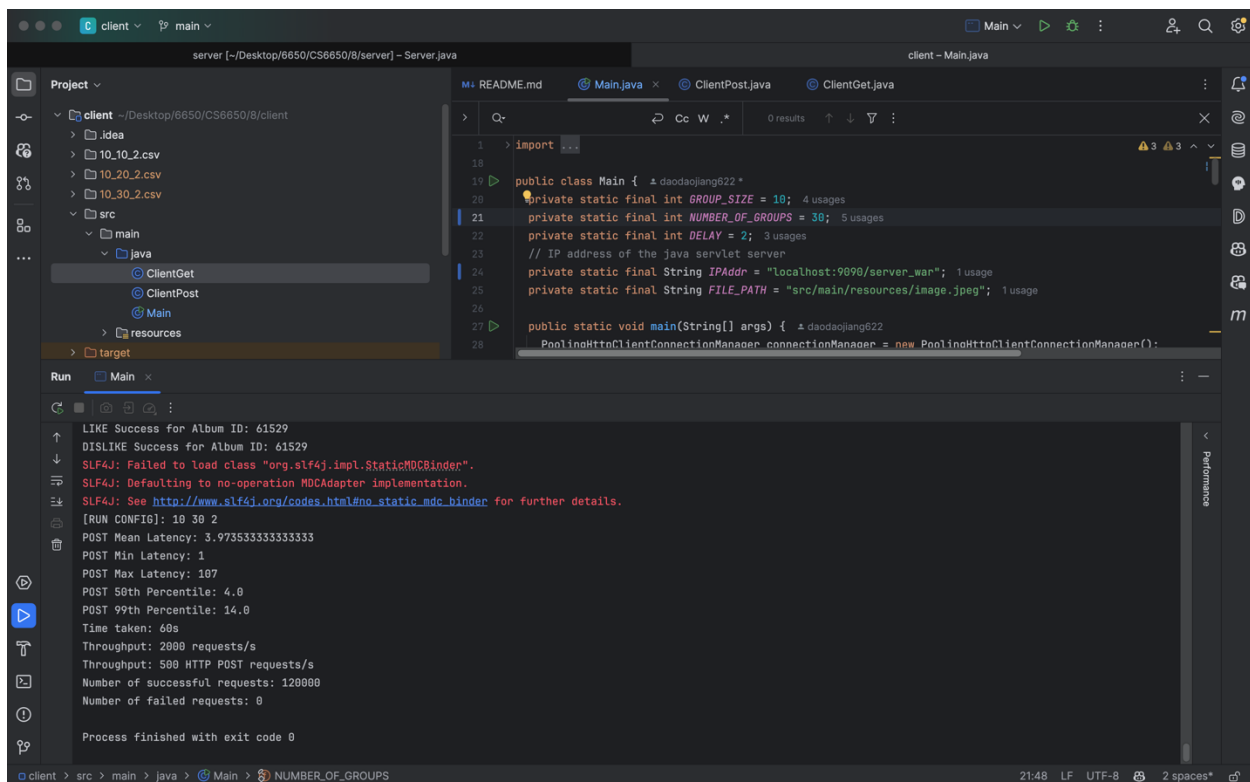
Project [~/Desktop/6650/CS6650/8/client]
  client
    .idea
    10_10_2.csv
    src
      main
        java
          ClientGet
          ClientPost
          Main

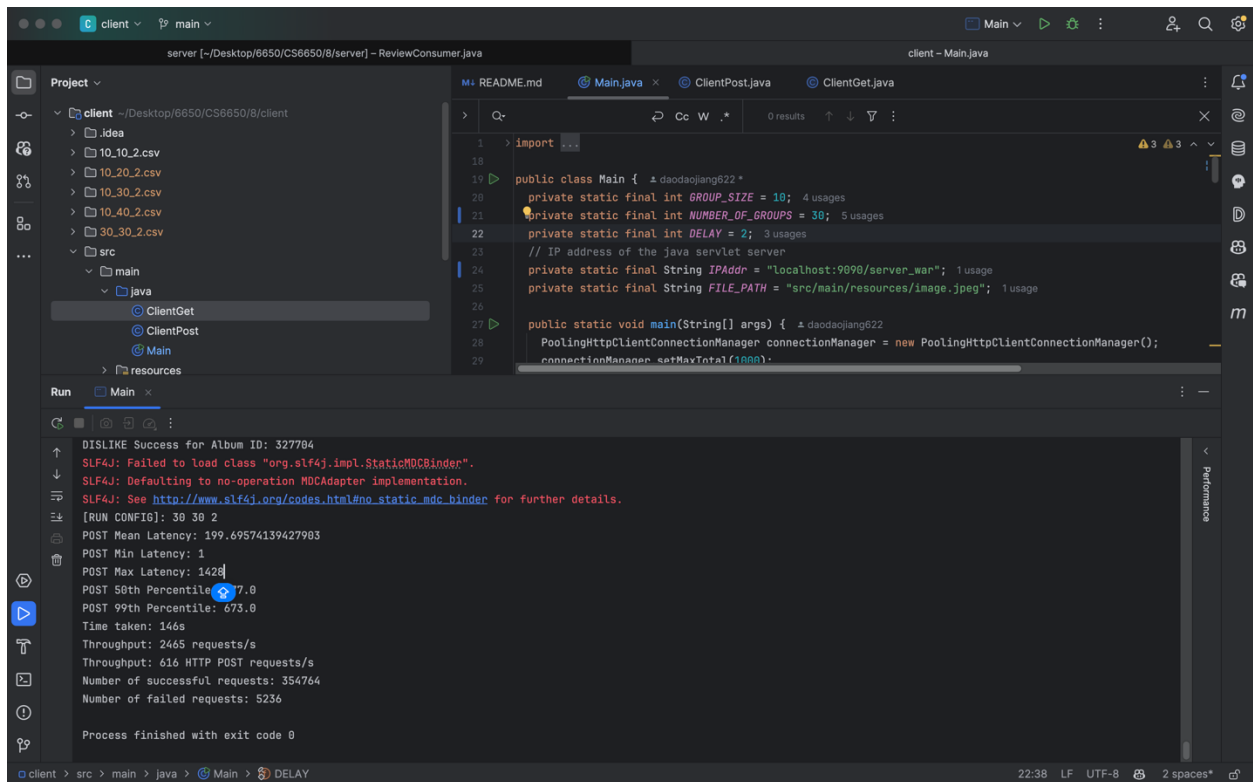
Run [Main]
  LIKE Success for Album ID: 11528
  LIKE Success for Album ID: 11528
  DISLIKE Success for Album ID: 11528
  Created Album ID: 11529
  LIKE Success for Album ID: 11529
  LIKE Success for Album ID: 11529
  DISLIKE Success for Album ID: 11529
  SLF4J: Failed to load class "org.slf4j.impl.StaticMDCBinder".
  SLF4J: Defaulting to no-operation MDCAdapter implementation.
  SLF4J: See http://www.slf4j.org/codes.html#no_static_mdc_binder for further details.
  [RUN CONFIG]: 10 10 2
  POST Mean Latency: 4.0708
  POST Min Latency: 0
  POST Max Latency: 186
  POST 50th Percentile: 4.0
  POST 99th Percentile: 14.0
  Time taken: 19s
  Throughput: 2105 requests/s
  Throughput: 526 HTTP POST requests/s
  Number of successful requests: 40000
  Number of failed requests: 0
  Process finished with exit code 0
```

```
server [~/Desktop/6650/CS6650/8/server] - Server.java
client - Main.java

Project [~/Desktop/6650/CS6650/8/client]
  client
    .idea
    10_10_2.csv
    src
      main
        java
          ClientGet
          ClientPost
          Main
        resources
        target
        .gitignore

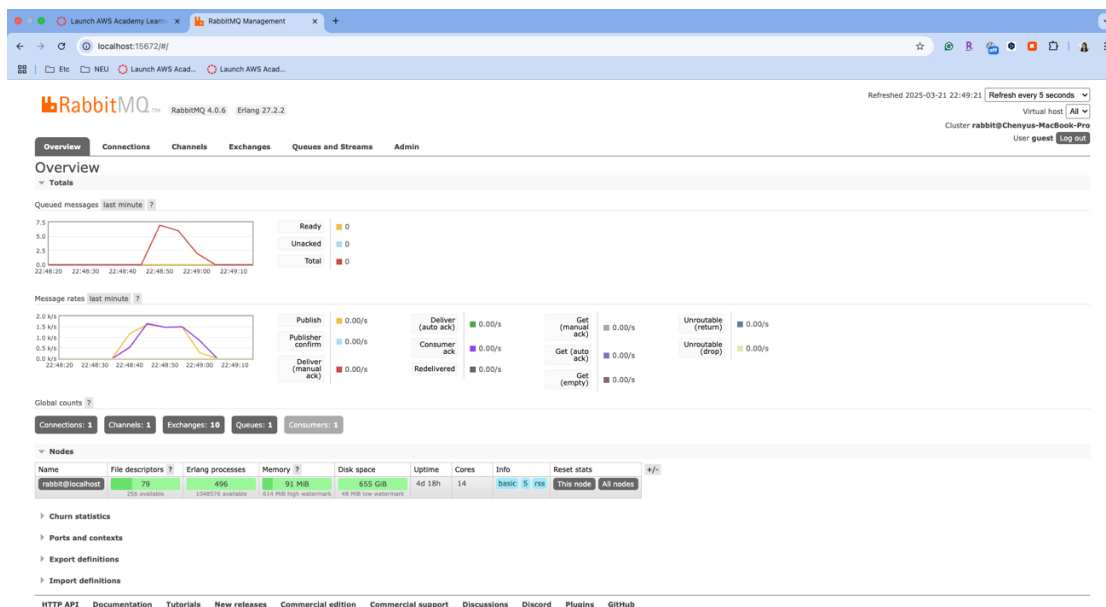
Run [Main]
  LIKE Success for Album ID: 31529
  DISLIKE Success for Album ID: 31529
  SLF4J: Failed to load class "org.slf4j.impl.StaticMDCBinder".
  SLF4J: Defaulting to no-operation MDCAdapter implementation.
  SLF4J: See http://www.slf4j.org/codes.html#no_static_mdc_binder for further details.
  [RUN CONFIG]: 10 20 2
  POST Mean Latency: 3.6899125
  POST Min Latency: 1
  POST Max Latency: 86
  POST 50th Percentile: 3.0
  POST 99th Percentile: 13.0
  Time taken: 39s
  Throughput: 2051 requests/s
  Throughput: 512 HTTP POST requests/s
  Number of successful requests: 80000
  Number of failed requests: 0
  Process finished with exit code 0
```



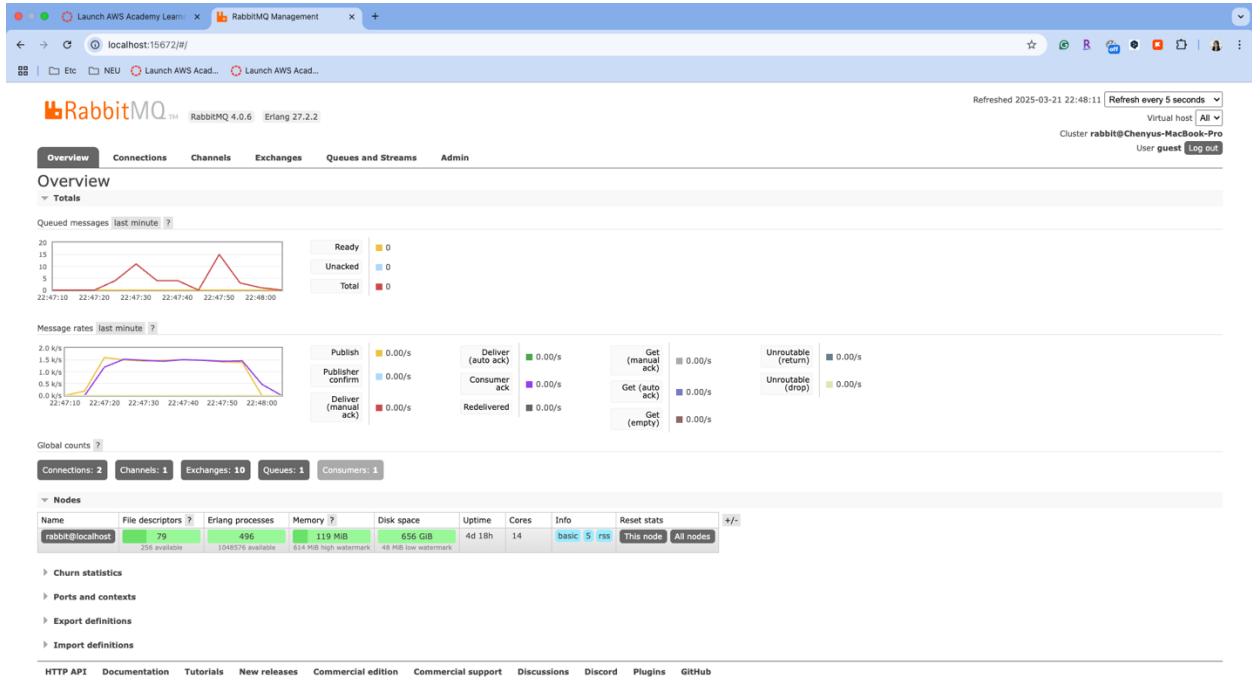


Queue sizes for 10 10 2, 10 20 2, and 10 30 2 never near 1000, indicating the current implementation of consumer is efficient enough to handle the threads. However, we can see it reaches 10k for 30 30 2, suggesting a need to a need to scale the consumer horizontally. The shapes of queue size have some plateau but are more closely resembles to pointy shape, also suggesting a need to scale horizontally.

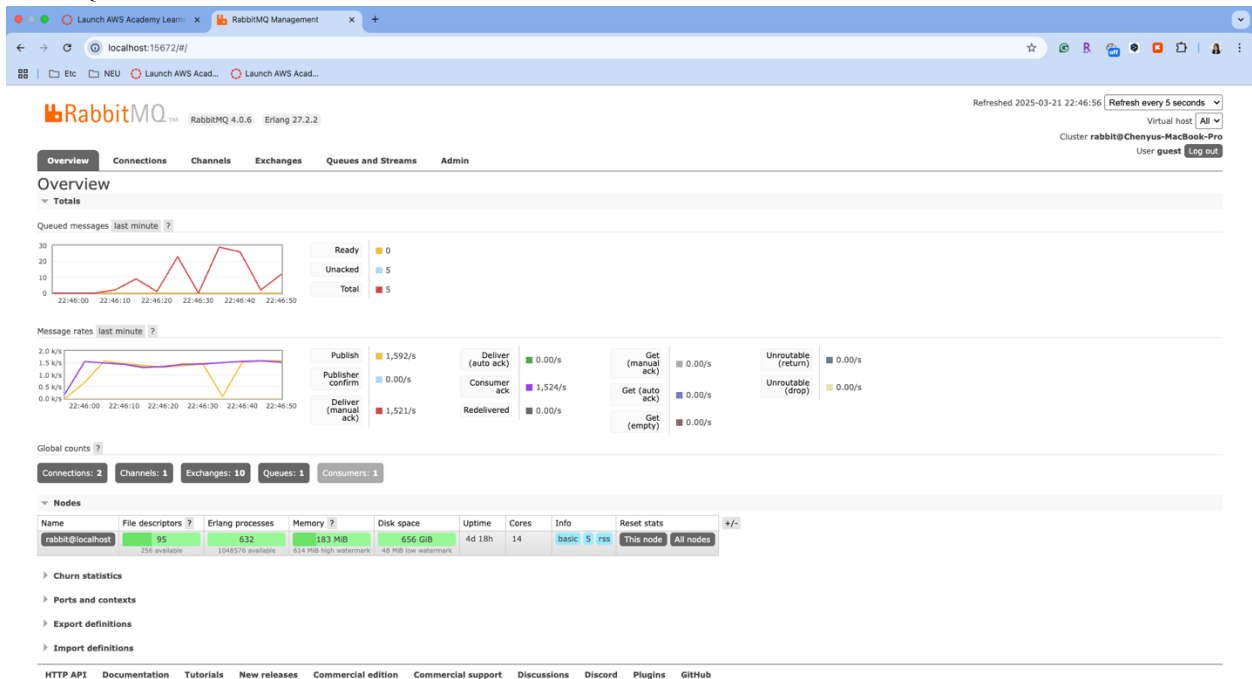
RabbitMQ statistics for 10 10 2:



RabbitMQ statistics for 10 20 2:



RabbitMQ statistics for 10 30 2:



RabbitMQ statistics for 30 30 2:

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localhost:15672/#/

RabbitMQ 4.0.6 Erlang 27.2.2

Overview Connections Channels Exchanges Queues and Streams Admin

Overview

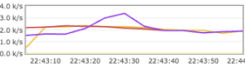
Totals

Queued messages [last minute](#) ?



Ready 0
Unacked 28
Total 28

Message rates [last minute](#) ?



Global counts ?

Connections: 3 Channels: 1 Exchanges: 10 Queues: 1 Consumers: 1

Nodes

Name	File descriptors ?	Erlang processes	Memory ?	Disk space	Uptime	Cores	Info	Reset stats
rabbit@localhost	196 25% available	1264 1048576 available	207 MB 22 GiB high watermark	659 GiB 48 MiB low watermark	4d 18h	14	basic 5 rxn	This node All nodes

Churn statistics

Ports and contexts

Export definitions

Import definitions

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